

■ 2.3.5 Finding Expressions that Match a Pattern

When you have generated an expression, especially a large one, you often want to pick out just those pieces that have a particular form. You can describe the kind of pieces you want by setting up a pattern. Then you can use the following functions to find the pieces that match the pattern.

<code>Cases[expr, form]</code>	give the elements in <i>expr</i> that match <i>form</i>
<code>Count[expr, form]</code>	give the number of elements in <i>expr</i> that match <i>form</i>
<code>Select[expr, test]</code>	give the elements of <i>expr</i> on which <i>test</i> gives True
<code>Position[expr, form]</code>	give a list of the positions of parts anywhere in <i>expr</i> that match <i>form</i>

Functions for finding parts of expressions.

Here is a list.

```
In[1]:= t = {3, x, 4, x^2, x^3, 5}
```

```
Out[1]= {3, x, 4, x^2, x^3, 5}
```

This picks out the elements of the list *t* that have the form of *x* to a power.

```
In[2]:= Cases[t, x^_]
```

```
Out[2]= {x^2, x^3}
```

This counts the number of elements in *t* that match *x^_*, without explicitly returning a list of them.

```
In[3]:= Count[t, x^_]
```

```
Out[3]= 2
```

This picks out the integers in *t*.

```
In[4]:= Cases[t, _Integer]
```

```
Out[4]= {3, 4, 5}
```

You can also use `Select` to pick out the integers.

```
In[5]:= Select[t, IntegerQ]
```

```
Out[5]= {3, 4, 5}
```

This gives two positions at which 3 occurs in *t*. The second 3 is in the exponent of *x*³.

```
In[6]:= Position[t, 3]
```

```
Out[6]= {{1}, {5, 2}}
```

This gives all terms involving *x*² in the polynomial.

```
In[7]:= Cases[Expand[(1 + x + y)^3], n_. x^2]
```

```
Out[7]= {3 x^2, 3 x^2 y}
```